

Table 12: Full Evaluation Results on Select. (Best F1 scores in bold, second-best in underline for each operand.)

Operand	Impl	GPT-5				Qwen3-Max				Claude-4.5-Sonnet			
		Precision	Recall	F1	Cost (\$)	Precision	Recall	F1	Cost (\$)	Precision	Recall	F1	Cost (\$)
Row-wise Select	LLM-ALL	0.946	0.946	0.938	0.004	0.812	0.918	0.826	0.003	0.801	0.877	0.807	0.006
	+CoT	0.947	0.957	0.945	0.005	0.946	0.909	0.917	0.004	0.851	0.881	0.846	0.009
	LLM-ONE	0.934	0.951	0.938	0.008	0.744	0.934	0.798	0.008	0.682	0.789	0.722	0.017
	+CoT	0.936	0.953	<u>0.940</u>	0.025	0.917	0.962	0.932	0.026	0.931	0.961	0.937	0.078
Column-wise Select	LLM-ALL	0.942	0.873	0.880	0.001	0.941	0.819	0.835	0.001	0.925	0.831	0.850	0.001
	+CoT	0.938	0.895	<u>0.891</u>	0.001	0.955	0.873	0.890	0.001	0.900	0.896	0.856	0.003
	+EX	0.947	0.841	<u>0.851</u>	0.001	0.941	0.906	0.903	0.001	0.941	0.802	0.826	0.003
	+CoT+EX	0.938	0.895	<u>0.891</u>	0.002	0.966	0.840	0.874	0.002	0.931	0.856	0.865	0.005
	LLM-ONE	0.857	0.801	0.811	0.003	0.840	0.903	0.839	0.003	0.766	0.870	0.794	0.008
	+CoT	0.819	0.819	0.805	0.012	0.779	0.918	0.816	0.013	0.791	0.857	0.810	0.040
	+EX	0.845	0.921	0.872	0.005	0.866	0.896	0.854	0.005	0.782	0.932	0.830	0.011
	+CoT+EX	0.860	0.893	0.849	0.014	0.803	0.994	0.869	0.015	0.781	0.820	0.771	0.043
Table-wise Select	LLM-ALL	1.000	0.917	0.944	0.001	0.929	0.955	0.928	0.001	0.949	0.976	<u>0.953</u>	0.003
	+CoT	0.983	0.952	0.963	0.002	0.962	0.961	0.951	0.002	0.921	0.983	0.943	0.005
	+EX	0.914	0.928	0.904	0.009	0.929	0.955	0.928	0.009	0.937	0.943	0.930	0.021
	+CoT+EX	0.946	0.919	0.921	0.009	0.945	0.957	0.939	0.010	0.921	0.983	0.943	0.024
	LLM-ONE	0.721	0.578	0.616	0.003	0.594	0.531	0.528	0.003	0.637	0.691	0.624	0.007
	+CoT	0.667	0.533	0.577	0.012	0.681	0.567	0.588	0.013	0.661	0.672	0.662	0.040
	+EX	0.725	0.678	0.670	0.011	0.674	0.604	0.606	0.011	0.646	0.716	0.664	0.025
	+CoT+EX	0.689	0.678	0.654	0.020	0.675	0.613	0.612	0.021	0.612	0.664	0.617	0.060

Table 13: Full Evaluation Results on Match. (Best F1 scores in bold, second-best in underline for each operand.)

LRO	Impl	GPT-5				Qwen3-Max				Claude-4.5-Sonnet			
		Precision	Recall	F1	Cost (\$)	Precision	Recall	F1	Cost (\$)	Precision	Recall	F1	Cost (\$)
Cell-wise Match	LLM-ALL	0.854	0.891	<u>0.838</u>	0.004	0.767	0.742	0.715	0.003	0.757	0.807	0.757	0.006
	+CoT	0.858	0.880	0.843	0.004	0.839	0.783	0.758	0.003	0.836	0.871	0.821	0.008
	LLM-ONE	0.843	0.851	0.818	0.045	0.623	0.766	0.627	0.042	0.527	0.727	0.564	0.092
	+CoT	0.833	0.837	0.802	0.174	0.709	0.802	0.704	0.185	0.761	0.820	0.753	0.513
	LLM-SEMI	0.829	0.766	0.752	0.006	0.556	0.715	0.565	0.006	0.585	0.740	0.603	0.020
	+CoT	0.834	0.823	0.789	0.014	0.736	0.789	0.729	0.014	0.777	0.816	0.756	0.047
Row-wise Match	LLM-ALL	0.975	0.956	0.961	0.027	0.804	0.792	0.795	0.019	0.949	0.951	<u>0.948</u>	0.052
	+CoT	0.975	0.956	0.961	0.029	0.982	0.677	0.735	0.024	0.922	0.948	0.929	0.058
	LLM-ONE	0.846	0.822	0.829	0.211	0.874	0.839	0.844	0.190	0.959	0.839	0.869	0.505
	+CoT	0.846	0.822	0.829	0.338	0.886	0.854	0.862	0.335	0.846	0.774	0.797	0.946
	LLM-SEMI	0.949	0.929	0.934	0.074	0.658	0.751	0.690	0.070	0.906	0.929	0.914	0.161
	+CoT	0.949	0.929	0.934	0.085	0.916	0.945	0.922	0.079	0.944	0.951	0.941	0.200
Column-wise Match	LLM-ALL	0.565	0.693	0.605	0.001	0.449	0.535	0.477	0.001	0.606	0.661	0.626	0.002
	+CoT	0.675	0.849	0.736	0.002	0.673	0.726	0.680	0.002	0.720	0.769	0.731	0.004
	+EX	0.670	0.855	<u>0.737</u>	0.003	0.662	0.746	0.693	0.002	0.373	0.351	0.360	0.006
	+CoT+EX	0.685	0.846	0.746	0.003	0.736	0.771	0.735	0.003	0.702	0.772	0.726	0.008
	LLM-ONE	0.490	0.569	0.506	0.021	0.589	0.520	0.516	0.023	0.130	0.431	0.189	0.014
	+CoT	0.600	0.637	0.588	0.054	0.483	0.629	0.519	0.124	0.467	0.790	0.553	0.361
	+EX	0.564	0.712	0.615	0.054	0.518	0.512	0.479	0.053	0.591	0.779	0.652	0.127
	+CoT+EX	0.537	0.700	0.591	0.154	0.393	0.526	0.415	0.166	0.491	0.723	0.571	0.484
	LLM-SEMI	0.602	0.863	0.695	0.002	0.120	0.264	0.161	0.002	0.479	0.814	0.578	0.008
	+CoT	0.586	0.846	0.680	0.007	0.261	0.458	0.326	0.007	0.468	0.747	0.563	0.021
	+EX	0.617	0.858	0.709	0.012	0.403	0.651	0.485	0.012	0.477	0.798	0.574	0.031
	+CoT+EX	0.619	0.852	0.705	0.017	0.533	0.779	0.623	0.019	0.546	0.802	0.637	0.050

Table 14: Full Evaluation Results on Impute. (Best LLM judge scores in bold, second-best in underline for each operand.)

Operand	Impl	GPT-5			Qwen3-Max			Claude-4.5-Sonnet		
		EM	LLM Judge Score	Cost (\$)	EM	LLM Judge Score	Cost (\$)	EM	LLM Judge Score	Cost (\$)
Cell-wise Impute	LLM-ALL	0.733	0.890	0.003	0.643	0.826	0.003	0.661	0.864	0.007
	+CoT	0.739	<u>0.886</u>	0.003	0.648	0.820	0.003	0.665	0.864	0.008
	LLM-ONE	0.473	0.681	0.001	0.427	0.634	0.001	0.423	0.643	0.004
	+CoT	0.476	0.686	0.002	0.412	0.673	0.003	0.433	0.654	0.008
	+EX	0.691	0.836	0.001	0.576	0.746	0.001	0.635	0.792	0.004
	+CoT+EX	0.674	0.848	0.002	0.596	0.824	0.003	0.634	0.833	0.008
Column-wise Impute	LLM-ALL	0.606	0.807	0.001	0.557	0.740	0.001	0.606	0.817	0.002
	+CoT	0.577	0.816	0.001	0.526	0.737	0.001	0.610	0.834	0.003
	LLM-ONE	0.618	0.826	0.002	0.659	0.805	0.002	0.644	0.812	0.005
	+CoT	0.640	<u>0.838</u>	0.005	0.621	0.793	0.006	0.657	0.859	0.017
Row-wise Impute	LLM-ONE	0.438	0.632	7.0×10^{-5}	0.447	0.644	8.0×10^{-5}	0.455	0.580	1.9×10^{-4}
	+CoT	0.422	0.607	1.5×10^{-4}	0.479	0.664	1.9×10^{-4}	0.433	0.561	4.8×10^{-4}
	+EX	0.674	<u>0.754</u>	7.0×10^{-5}	0.667	0.764	7.0×10^{-5}	0.664	0.719	1.9×10^{-4}
	+CoT+EX	0.674	0.724	1.5×10^{-4}	0.662	0.751	2.1×10^{-4}	0.659	0.714	5.2×10^{-4}

Table 15: Full Evaluation Results on Cluster. (Best ARI, NMI in bold, second-best in underline for each operand.)

Operand	Impl	GPT-5			Qwen3-Max			Claude-4.5-Sonnet		
		ARI	NMI	Cost (\$)	ARI	NMI	Cost (\$)	ARI	NMI	Cost (\$)
Row-wise Cluster	LLM-ALL	0.736	0.809	0.007	<u>0.731</u>	<u>0.806</u>	0.005	0.674	0.768	0.013
	+CoT	0.723	0.801	0.008	0.720	0.778	0.006	0.675	0.763	0.016
	LLM-ONE	0.694	0.754	0.017	0.684	0.746	0.015	0.688	0.757	0.038
	+CoT	0.694	0.759	0.029	0.658	0.727	0.026	0.696	0.756	0.073
Column-wise Cluster	LLM-ALL	0.804	0.872	0.008	0.794	0.855	0.005	0.807	0.874	0.012
	+CoT	0.787	0.850	0.008	0.776	0.841	0.006	0.784	0.845	0.014
	+EX	0.822	0.890	0.009	0.807	0.869	0.007	<u>0.843</u>	0.900	0.016
	+CoT+EX	0.812	0.877	0.010	0.751	0.835	0.008	0.848	<u>0.897</u>	0.018
	LLM-ONE	0.611	0.715	0.015	0.628	0.742	0.013	0.661	0.744	0.033
	+CoT	0.616	0.724	0.026	0.578	0.692	0.024	0.638	0.733	0.068
	+EX	0.598	0.720	0.017	0.549	0.694	0.015	0.640	0.735	0.037
	+CoT+EX	0.604	0.721	0.028	0.493	0.641	0.027	0.610	0.726	0.077
Table-wise Cluster	LLM-ALL	<u>0.760</u>	<u>0.865</u>	0.004	0.771	0.869	0.003	0.744	0.845	0.007
	+CoT	0.702	0.831	0.005	0.659	0.800	0.004	0.704	0.830	0.009
	+EX	0.699	0.836	0.015	0.743	0.852	0.014	0.707	0.831	0.033
	+CoT+EX	0.721	0.837	0.015	0.627	0.783	0.015	0.716	0.830	0.035
	LLM-ONE	0.615	0.755	0.006	0.509	0.692	0.006	0.629	0.767	0.014
	+CoT	0.669	0.792	0.010	0.538	0.721	0.011	0.613	0.749	0.029
	+EX	0.679	0.810	0.017	0.626	0.782	0.016	0.623	0.771	0.040
	+CoT+EX	0.668	0.801	0.023	0.630	0.780	0.023	0.647	0.788	0.059

Table 16: Full Evaluation Results on Order. (Best HR@k, Kendall τ in bold, second-best in underline.)

Operand	Impl	GPT-5			Qwen3-Max			Claude-4.5-Sonnet		
		HR@k	Kendall τ	Cost (\$)	HR@k	Kendall τ	Cost (\$)	HR@k	Kendall τ	Cost (\$)
Row-wise Order	LLM-ALL	0.929	0.899	0.001	0.861	0.693	0.001	<u>0.945</u>	0.848	0.003
	+CoT	0.941	0.937	0.002	0.952	0.884	0.002	0.949	0.865	0.006
	LLM-PAIR	0.929	0.905	0.067	0.901	0.757	0.067	0.767	0.688	0.133
	+CoT	0.929	0.907	0.182	0.921	0.892	0.208	0.910	<u>0.914</u>	0.568
	LLM-SORT	0.885	0.867	0.008	0.901	0.858	0.011	0.828	<u>0.816</u>	0.024
	+CoT	0.896	0.881	0.021	0.941	0.884	0.033	0.899	0.871	0.083
	LLM-SCORE	0.789	0.690	0.004	0.831	0.686	0.004	0.807	0.707	0.009
	+CoT	0.776	0.658	0.011	0.805	0.588	0.014	0.798	0.706	0.039